

Gradient Boosting Regressor

House	Size	Bedrooms	Age	Actual price
H1	1000	2	20	50
H2	1200	2	15	55
H3	1500	3	10	70
H4	1800	3	5	85

Target = Price

Core idea of gradient Boosting

Build small tree

Find mistakes (error, residual)

Build another tree to fix mistakes

Repeat

$$F(x) = T_1(x) + T_2(x) + T_3(x) \dots$$

Each new tree learns

where previous tree were wrong

Step 1

First prediction

$$\text{Mean price} = (50 + 55 + 70 + 85) / 4 = 65$$

So initial model predicts 65 for every row

Prediction after first model

House	Actual	predicted
H1	50	65
H2	55	65
H3	70	65
H4	85	65

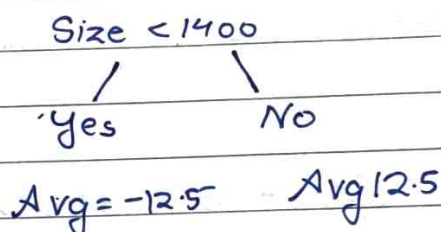
Step 2: Calculate Residual

House	Actual	Predicted	Residual
H ₁	50	65	-15
H ₂	55	65	-10
H ₃	70	65	+5
H ₄	85	65	+20

Step 3: Train tree on Residual

Now we train small decision tree to predict
Targets = [-15, -10, +5, +20]

Best split



Left side

$$H_1 = -15$$

$$H_2 = -10$$

$$\text{Avg} = -12.5$$

Right side

$$H_3 = 5$$

$$H_4 = 20$$

$$\text{Avg} = 12.5$$

Step 4: Update prediction

$$F_1(x) = F_0(x) + \eta T_1(x)$$

$$\eta = 0.1$$

$$H_1 = 65 + 0.1(-12.5) = 63.5$$

$$H_2 = 63.5 \text{ (same)}$$

$$H_3 \& H_4 = 65 + 0.1(12.5) = 66.25$$

Prediction After Tree 1

House	Actual	New prediction
H ₁	50	63.75
H ₂	55	63.75
H ₃	70	66.25
H ₄	85	66.25

Note: prediction moves towards correct value

Calculate new residual

create tree 2

Repeat the process

Points to remember

- it is called gradient boosting bcoz it depends on $(y - \hat{y})^2$

$$\frac{\partial}{\partial y} (y - \hat{y})^2 = -2(y - \hat{y})$$

Residual are proportional to -ve gradient
 new tree learns direction of gradient

- Tree depth usually = 3-8
- Learning rate lower better generalization
- More tree better fitting, but can lead to overfitting
- Sequential learning